

1. (35 points) Find the solution set of

$$\begin{aligned}x_1 + 2x_2 + 2x_3 + x_4 &= 3 \\ -2x_1 - 4x_2 - 3x_3 - x_4 &= -5 \\ x_1 + 2x_2 + x_3 &= 2\end{aligned}$$

using Gauss-Jordan elimination.

Augmented matrix of the per system is:

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 1 & 3 \\ -2 & -4 & -3 & -1 & -5 \\ 1 & 2 & 1 & 0 & 2 \end{array} \right] \xrightarrow{\substack{2R_1 + R_2 \rightarrow R_2 \\ -R_1 + R_3 \rightarrow R_3}} \left[\begin{array}{cccc|c} 1 & 2 & 2 & 1 & 3 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & -1 & -1 & -1 \end{array} \right] \xrightarrow{-R_3 \rightarrow R_3}$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 1 & 3 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{array} \right] \xrightarrow{-R_2 + R_3 \rightarrow R_3} \left[\begin{array}{cccc|c} 1 & 2 & 2 & 1 & 3 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{-2R_2 + R_1 \rightarrow R_1}$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{\substack{x_3 + x_4 = 1 \\ x_1 + 2x_2 - x_4 = 1}} \text{So, let } \begin{cases} x_4 = \alpha \\ x_2 = \beta \end{cases}$$

$$\rightarrow (x_1, x_2, x_3, x_4) = (1 + \alpha - 2\beta, \beta, 1 - \alpha, \alpha)$$

$$\begin{cases} \alpha \in \mathbb{R} \\ \beta \in \mathbb{R} \end{cases}$$

2. (30 points) Let $z \in \mathbb{C}$. If $|z| = 2$, prove that

$$|z^4 + z^2 - 2| \geq 6$$

$$|z^4 + z^2 - 2| = |(z^2 + 2)(z^2 - 1)|$$

$$= |z^2 + 2| |z^2 - 1|$$

by triangle
inequality
on the problem
set questions.

$$\geq (|z^2| - |2|)(|z^2| - |1|)$$

$$= (|z|^2 - 2)(|z|^2 - 1)$$

$$= 2 \cdot 3 = 6$$

$$\Rightarrow |z^4 + z^2 - 2| \geq 6$$

3. (35 points) Determine whether $\mathbb{R}^3$ is a vector space over $\mathbb{R}$ with the vector addition $\oplus$ defined as:

$$(a_1, a_2, a_3) \oplus (b_1, b_2, b_3) = (a_1 + b_1 + 5, a_2 + b_2 - 7, a_3 + b_3 + 1)$$

and the scalar multiplication $\odot$ defined as:

$$c \odot (a_1, a_2, a_3) = (ca_1 + 5(c-1), ca_2 - 7(c-1), ca_3 + c - 1).$$

If not, give at least one property that is not satisfied.

$$\begin{aligned} \textcircled{i)} (a_1, a_2, a_3) \oplus (b_1, b_2, b_3) &= (a_1 + b_1 + 5, a_2 + b_2 - 7, a_3 + b_3 + 1) \\ &= (b_1 + a_1 + 5, b_2 + a_2 - 7, b_3 + a_3 + 1) \\ &= (b_1, b_2, b_3) \oplus (a_1, a_2, a_3) \end{aligned}$$

$$\begin{aligned} \textcircled{ii)} [(a_1, a_2, a_3) \oplus (b_1, b_2, b_3)] \oplus (c_1, c_2, c_3) &= (a_1 + b_1 + 5, a_2 + b_2 - 7, a_3 + b_3 + 1) \oplus (c_1, c_2, c_3) \\ &= (a_1 + b_1 + 5 + c_1 + 5, a_2 + b_2 - 7 + c_2 - 7, a_3 + b_3 + 1 + c_3 + 1) \\ &= (a_1 + (b_1 + c_1 + 5) + 5, a_2 + (b_2 + c_2 - 7) - 7, a_3 + (b_3 + c_3 + 1) + 1) \\ &= (a_1, a_2, a_3) \oplus (b_1 + c_1 + 5, b_2 + c_2 - 7, b_3 + c_3 + 1) \\ &= (a_1, a_2, a_3) \oplus [(b_1, b_2, b_3) \oplus (c_1, c_2, c_3)] \end{aligned}$$

$$\textcircled{iii)} \forall (a_1, a_2, a_3) \in \mathbb{R}^3,$$

$$(a_1, a_2, a_3) \oplus (-5, 7, -1) = (-5, 7, -1) \oplus (a_1, a_2, a_3) = (a_1, a_2, a_3)$$

$\Rightarrow (-5, 7, -1)$ is the additive identity.

$$\textcircled{iv)} \forall (a_1, a_2, a_3) \in \mathbb{R}^3,$$

$$(a_1, a_2, a_3) \oplus (-a_1 - 10, -a_2 + 14, -a_3 - 2) = (-5, 7, -1)$$

$\Rightarrow (-a_1 - 10, -a_2 + 14, -a_3 - 2)$ is the additive inverse of (a_1, a_2, a_3)

$$\begin{aligned}
 (2)(i) (c+d) \odot (a_1, a_2, a_3) &= ((c+d)a_1 + 5(c+d-1), (c+d)a_2 - 7(c+d-1), (c+d)a_3 + c+d-1) \\
 &= (ca_1 + da_1 + 5c + 5d - 5, ca_2 + da_2 - 7c - 7d + 7, ca_3 + da_3 + c + d - 1) \\
 &= ((ca_1 + 5(c-1)) + (da_1 + 5(d-1)) + 5, (ca_2 - 7(c-1)) + (da_2 - 7(d-1)) + 7, \\
 &\quad (ca_3 + c - 1) + (da_3 + d - 1) + 1) \\
 &= [c \odot (a_1, a_2, a_3)] \oplus [d \odot (a_1, a_2, a_3)]
 \end{aligned}$$

$$(ii) (cd) \odot (a_1, a_2, a_3) = (cd a_1 + 5(cd-1), (cd) a_2 - 7(cd-1), (cd) a_3 + cd-1) \quad (*)$$

on the other hand,

$$\begin{aligned}
 c \odot (d \odot (a_1, a_2, a_3)) &= c \odot (da_1 + 5(d-1), da_2 - 7(d-1), da_3 + d-1) \\
 &= (cd a_1 + 5(c(d-1)) + 5(c-1), cd a_2 - 7c(d-1) - 7(c-1), cd a_3 + cd - c + c - 1) \\
 &= (cd a_1 + 5(cd-1), cd a_2 - 7(cd-1), cd a_3 + cd - 1) \quad (***)
 \end{aligned}$$

(*) = (***) the associativity is satisfied

$$\begin{aligned}
 (iii) c \odot [(a_1, a_2, a_3) \oplus (b_1, b_2, b_3)] &= c \odot [(a_1 + b_1 + 5, a_2 + b_2 - 7, a_3 + b_3 + 1)] \\
 &= [c(a_1 + b_1 + 5) + 5(c-1), c(a_2 + b_2 - 7) - 7(c-1), c(a_3 + b_3 + 1) + c - 1] \\
 &= [(ca_1 + 5(c-1)) + (cb_1 + 5(c-1)) + 5, (ca_2 - 7(c-1)) + (cb_2 - 7(c-1)) + 7, \\
 &\quad (ca_3 + c - 1) + (cb_3 + c - 1) + 1]
 \end{aligned}$$

$$= (ca_1 + 5(c-1), ca_2 - 7(c-1), ca_3 + c - 1) \oplus$$

$$(cb_1 + 5(c-1), cb_2 - 7(c-1), cb_3 + c - 1)$$

$$= [c \odot (a_1, a_2, a_3)] \oplus [c \odot (b_1, b_2, b_3)]$$

$$(iv) 1 \odot (a_1, a_2, a_3) = (a_1, a_2, a_3)$$

All of the 8 properties are satisfied, $\mathbb{R}^3$ is a v.s. with the given addition and scalar multiplication.